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Adaptor for providing an adjustable connection between an endpiece of a lifting apparatus and a nacelle, and method of using same

Abstract

An adaptor (166) for interfacing between an endpiece (38) of a lifting apparatus (30) and a nacelle (14) of a wind turbine (10). The adaptor (166) includes a base plate (174) having projections (204) configured to engage the adaptor (166) to the nacelle (14) and first and second side legs (176, 178) extending away from the base plate (174) opposite the projections (204) that define a recess (202) configured to engage the adaptor (166) with the endpiece (38) of the lifting apparatus (30). The adaptor (166) further includes first and second set screws (208, 210) positionable in bores (212) of the first and second side legs (176, 178) and movable to laterally adjust the position the adaptor (166) relative to the endpiece (38) of the lifting apparatus (30). This allows the lifting apparatus (30) to lift a nacelle assembly (24) having a laterally offset centre of gravity (158) in a balanced manner. A method for handling multiple wind turbine (10) components, including a nacelle (14), through selective use of the adaptor (166) is also disclosed.

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Background/Summary

TECHNICAL FIELD

(1) This invention relates generally to wind turbines, and more particularly to an adaptor disposed between an endpiece of a lifting apparatus and a transport frame of a nacelle assembly that provides an improved, tool-less connection to the transport frame and is adjustable to accommodate lateral displacements of the centre of gravity of the nacelle assembly, and to a method of using the adaptor to hoist a nacelle assembly in a balanced manner using the lifting apparatus.

BACKGROUND

(2) Wind turbine generators are used to produce electrical energy using a renewable resource and without combusting a fossil fuel. A wind turbine generator converts kinetic energy from the wind into electrical energy, and includes a tower, a nacelle mounted atop the tower, a rotor hub rotatably supported by the nacelle, and a plurality of rotor blades attached to the hub. The hub is coupled to a generator housed inside the nacelle. Consequently, as wind forces the blades to rotate, electrical energy is produced by the generator.

(3) In a typical installation process, various components of the wind turbine, such as the nacelle, tower or tower sections, rotor hub, rotor blades, etc., are manufactured separately at one or more manufacturing facilities and then transported to an installation site for assembly. This process requires that the various components be loaded onto and unloaded from transportation means, which may include trucks, railroad cars, ships, and/or other suitable transportation vehicles. In this regard, the manufacturing facility or an intermediate loading facility (e.g., quayside for offshore wind turbines) may include various lifting apparatus for moving the wind turbine components around the facility and/or to their desired location on the transportation means.

(4) By way of example, one or more cranes may be used at a facility or location to move the components about the facility or load/unload wind turbine components to/from a transportation vehicle. More particularly, a dual-crane system may be used to move tower sections at a facility or load/unload location. In this system, a first crane may connect to and hoist a first end of the tower section and a second crane may connect to and hoist a second end of the tower section. The first and second cranes may then coordinate their movements to move the tower section to a desired location. To lift the ends of the tower sections, the cranes include an endpiece that facilitates a coupling with the respective ends of the tower section. By way of example, the endpiece may be configured as a J-hook for attaching to the ends of the tower sections, as is generally known in the art.

(5) In a dual-crane system, J-hooks are particularly well suited for lifting loads that have a regular geometry and a predictable centre of gravity. More particularly, the centre of gravity of a load being lifted by J-hooks in a dual-crane system should preferably be within a vertical geometrically-centered plane that includes the J-hooks. A lateral offset of the centre of gravity from this central plane will cause the load to tilt when it is lifted by the dual-crane system. Because tower sections have a regular geometry and a centre of gravity that generally lies along the geometric central plane of the tower section, J-hooks are extensively used for the crane endpiece when handling tower sections.

(6) One or more cranes, such as a dual-crane arrangement, may also be used at a facility or handling location to move nacelles. Unlike tower sections, however, nacelles may have a centre of gravity that does not lie along a geometric central plane of the nacelles, and the J-hooks used to handle the tower sections are thought to be unsuitable to move the nacelles. Accordingly, technicians or other personnel at the facility or handling location must then switch out the crane

endpieces and configure those endpieces to attach to the nacelle without damaging the nacelle. This switch out and connection process tends to be fairly complex and time-consuming to ensure a balanced lift of the nacelle with the crane(s) and without imposing undue stresses on the nacelle during the lift. Thus, the moving or loading/unloading process of wind turbine components with a crane arrangement is significantly slowed as a result of the cranes requiring different endpieces and/or connection means for handling different wind turbine components.

(7) In view of the above, manufacturers seek improved designs for handling multiple wind turbine components at a facility, load/unload location, etc. using lifting apparatus, such as a dual-crane arrangement, that avoids the need for switch-out processes and complex connections to accommodate the different components. Moreover, solutions are sought that allow a crane endpiece to be used to move different wind turbine components even when the components' centre of gravity may be slightly laterally offset from a geometric central plane of the components. Furthermore, manufacturers seek improved methods for handling different wind turbine components when their centre of gravity may be slightly laterally offset from the geometric central plane of the components.

SUMMARY

(8) An adaptor for interfacing between an endpiece of a lifting apparatus and a nacelle of a wind turbine includes a base plate having an outer surface and an inner surface, the outer surface having one or more projections configured to engage the adaptor to the nacelle; a first side leg extending away from the inner surface of the base plate, wherein the first side leg includes an outer surface, an inner surface, and a first bore extending between the inner and outer surfaces of the first side leg; a second side leg extending away from the inner surface of the base plate and spaced from the first side leg, wherein the second side leg includes an outer surface, an inner surface, and a second bore extending between the inner and outer surfaces of the second side leg. The base plate, first side leg, and second side leg collectively define a recess configured to engage the adaptor with the endpiece of the lifting apparatus. The adaptor further includes a first set screw positionable in the first bore such that an abutment end of the first set screw is disposed within the recess, and a second set screw positionable in the second bore such that an abutment end of the second set screw is disposed within the recess. The first and second set screws are configured to be movable, relative to the first and second bores, respectively, toward and away from each other to adjustably position the adaptor relative to the endpiece of the lifting apparatus.

(9) In one embodiment, the base plate is generally rectangular and may further include a front surface, a rear surface, and opposed side surfaces, wherein the first and second side legs extend from the inner surface adjacent respective side surfaces of the base plate. Additionally, the first and second side legs may extend from the inner surface adjacent to the rear surface of the base plate. Furthermore, the first and second side legs may have a length that is less than a length of the base plate. Moreover, in an exemplary embodiment, the first and second side legs may be aligned across a width of the base plate. Still further, the first and second side legs may extend from the inner surface of the base plate at a substantially right angle.

(10) In one embodiment, the first and second set screws and the first and second bores may be threaded such that rotation of the set screws moves the first and second set screws toward and away from each other. The first bore and the second bore may extend through their respective side legs in a direction substantially parallel to the base plate. Additionally, the first bore and the second bore may extend through their respective side legs adjacent the inner surface of the base plate. In one embodiment, the first and second set screws are movable toward and away from each other along an axis, and the adaptor is configured to be adjustable relative to the endpiece of the lifting apparatus in a direction substantially parallel to the axis.

(11) In an exemplary embodiment, the one or more projections may include a plurality of posts aligned and spaced apart on the upper surface of the base plate, wherein each of the plurality of posts has a tapered end. Furthermore, the adaptor may be configured to mount to the nacelle in a

tool-less manner. In an exemplary embodiment, the endpiece includes a J-hook and the nacelle includes a transport frame, wherein the adaptor is configured to be disposed between the J-hook and the transport frame attached to the nacelle.

(12) In another embodiment, an endpiece assembly for a lifting apparatus having a lifting cable includes an endpiece configured to be coupled to the lifting cable of the lifting apparatus. The endpiece defines a lateral direction and the adaptor, as described above, mounted to the endpiece. The position of the adaptor relative to the endpiece is adjustable in the lateral direction and is coupled to the endpiece at least by the first and second set screws. In an exemplary embodiment, the endpiece may be a J-hook. The J-hook may further include a first leg having an eyelet for attaching to the lifting cable of the lifting apparatus and a first edge, and a second leg extending from the first leg and having a second edge. The adaptor is mounted on the second edge such that the inner surface of the base plate engages the second edge and the second leg of the J-hook is disposed in the recess between first and second side legs of the adaptor. In one embodiment, the distance between the inner surfaces of the first and second side legs of the adaptor is greater than a thickness of the second leg of the J-hook to allow for adjustability of the adaptor in the lateral direction. The J-hook may additionally include a lock jaw for further securing the adaptor to the J-hook.

(13) In a further embodiment, a method of handling wind turbine components at a facility includes providing a lifting apparatus having a lifting cable and an endpiece on the lifting cable configured to interface with a load for lifting with the lifting apparatus. The method includes providing a first wind turbine component for moving with the lifting apparatus, the first wind turbine component having a centre of gravity laterally offset from a central plane of the first wind turbine component; selectively disposing an adaptor between the endpiece of the lifting apparatus and the first wind turbine component, the adaptor being adjustable relative to the endpiece of the lifting apparatus in a lateral direction; positioning the adaptor relative to the endpiece of the lifting apparatus so that the endpiece is vertically aligned with the offset centre of gravity of the first wind turbine component; and lifting the first wind turbine component with the lifting apparatus.

(14) In one embodiment, the adaptor between the endpiece of the lifting apparatus and the first wind turbine component includes attaching the adaptor to the endpiece of the lifting apparatus to form an endpiece assembly and subsequently attaching the endpiece assembly to the first wind turbine component. In an exemplary embodiment, the attachment of the endpiece assembly to the first wind turbine component may be tool-less.

(15) In one embodiment, positioning the adaptor relative to the endpiece of the lifting apparatus may include predetermining the lateral offset of the centre of gravity of the first wind turbine component and, prior to lifting the first wind turbine component, adjusting the position of the adaptor relative to the endpiece of the lifting apparatus to correspond to the predetermined lateral offset. Additionally or alternatively, positioning the adaptor relative to the endpiece of the lifting apparatus may include iteratively adjusting the position of the adaptor relative to the endpiece of the lifting apparatus and lifting the first wind turbine component with the lifting apparatus until the first wind turbine component fails to tilt when lifted by the lifting apparatus.

(16) In still a further embodiment, the method may include using the endpiece of the lifting apparatus without the adaptor to lift a second wind turbine component at the facility. In this way, the endpiece of the lifting apparatus may be used to handle two different wind turbine components at the facility without having to change out the endpiece of the lifting apparatus. Instead, the adaptor merely has to be inserted or removed in order to use the endpiece for the different wind turbine components.

(17) In one embodiment, the first wind turbine component may include a nacelle assembly including a nacelle having a front end and a rear end, a first transport frame attached to the front end of the nacelle, and a second transport frame attached to the rear end of the nacelle. The nacelle assembly has a laterally offset centre of gravity relative to its central plane. In one embodiment, the

second wind turbine component may include a tower section, where the centre of gravity is generally along the central plane of the tower section.

(18) In one embodiment, the lifting apparatus includes a dual-crane system having a first crane with a first lifting cable and a first endpiece and a second crane with a second lifting cable and a second endpiece. In this case, the method may include selectively disposing a first adaptor between the first endpiece and the first transport frame of the nacelle assembly; positioning the first adaptor relative to the first endpiece of the lifting apparatus so that the first endpiece is vertically aligned with the offset centre of gravity of the nacelle assembly; selectively disposing a second adaptor between the second endpiece and the second transport frame of the nacelle assembly; positioning the second adaptor relative to the second endpiece of the lifting apparatus so that the second endpiece is vertically aligned with the offset centre of gravity of the nacelle assembly; and lifting the nacelle assembly with the dual crane system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with a general description of the invention given above, and the detailed description given below, serve to explain the invention.

(2) FIG. 1 is a front perspective view of a wind turbine;

(3) FIG. 2 is a side elevation view of a nacelle assembly including a nacelle with transport frames on opposed ends;

(4) FIG. 3 is a side elevation view of a nacelle assembly being lifted by a lifting apparatus configured as a dual-crane system;

(5) FIG. 4 is a front perspective view of a transport frame;

(6) FIG. 5 is a rear perspective view of the transport frame shown in FIG. 4;

(7) FIG. 6 is a perspective view of an endpiece for a lifting apparatus configured as a J-hook;

(8) FIGS. 7A and 7B are schematic illustrations of lifting a nacelle assembly with a laterally offset centre of gravity with J-hooks;

(9) FIG. 8 is a disassembled perspective view of an adaptor in accordance with an embodiment of the invention;

(10) FIGS. 8A and 8B are side views of the adaptor illustrated in FIG. 8;

(11) FIGS. 8C and 8D are front and rear elevation views of the adaptor illustrated in FIG. 8;

(12) FIG. 9 is a perspective view of a J-hook assembly in accordance with an embodiment of the invention;

(13) FIGS. 10A and 10B are schematic illustrations of connecting the J-hook assembly shown in FIG. 9 to a transport frame of a nacelle assembly;

(14) FIG. 11 is a partial cross-sectional end view of the nacelle assembly illustrating a lateral offset of the adaptor relative to the J-hook for a balanced lift of a nacelle assembly; and

(15) FIG. 12 is a partial cross-sectional end view of the nacelle assembly illustrating the adaptor and J-hook during an iterative process to a balanced configuration.

DETAILED DESCRIPTION

(16) With reference to FIG. 1, a wind turbine 10 includes a tower 12, a nacelle 14 disposed at the apex of the tower 12, and a rotor 16 operatively coupled to a generator (not shown) via a gearbox (not shown) housed inside the nacelle 14. In addition to the generator and gearbox, the nacelle 14 may house various components needed to convert wind energy into electrical energy and to operate and optimize the performance of the wind turbine 10. The tower 12 supports the load presented by the nacelle 14, rotor 16, and other wind turbine components housed inside the nacelle 14 and operates to elevate the nacelle 14 and rotor 16 to a height above ground level or sea level, as may

be the case, at which air currents having lower turbulence and higher velocity are typically found.

(17) The rotor **16** may include a central hub **18** and a plurality of blades **20** attached to the central hub **18** at locations distributed about the circumference of the central hub **18**. In the representative embodiment, the rotor **16** includes three blades **20**, however the number may vary. The blades **20**, which project radially outward from the central hub **18**, are configured to interact with passing air currents to produce rotational forces that cause the rotor **16** to spin about its longitudinal axis. The design, construction, and operation of the blades **20** are familiar to a person having ordinary skill in the art of wind turbine design and may include additional functional aspects to optimize performance. For example, pitch angle control of the blades **20** may be implemented by a pitch control mechanism (not shown) responsive to wind velocity to optimize power production in low wind conditions, and to feather the blades if wind velocity exceeds design limitations. Wind exceeding a minimum level may activate the rotor **16**, causing the rotor **16** to rotate in a direction substantially perpendicular to the wind, applying torque to the input shaft of the generator. The electrical power produced by the generator may be supplied to a power grid (not shown) or an energy storage system (not shown) for later release to the grid as understood by a person having ordinary skill in the art. In this way, the kinetic energy of the wind may be harnessed by the wind turbine **10** for power generation.

(18) FIG. 2 illustrates a nacelle assembly **24** in accordance with an embodiment of the invention. The nacelle assembly **24** includes the nacelle **14** and a pair of transport frames **26** attached to front and rear ends of the nacelle **14**. The transport frames **26** are configured to facilitate the handling of a nacelle **14** for various purposes. For example, the transport frames **26** may aid in handling the nacelle **14** during manufacturing and/or storage of a nacelle **14** at a manufacturing facility. The transport frames **26** may also be used during the transport of a nacelle **14** from the manufacturing site to the installation site. In many instances, the transport frames **26** are attached to the nacelle **14** during manufacturing or shortly thereafter. In this way, the storage and/or handling of nacelles **14** at particular locations are often done in the context of the nacelle assembly **24**, as opposed to only the nacelle **14**. FIG. 2, for example, illustrates the nacelle assembly **24** positioned on a support surface **28** at a handling location during storage, for example. The support surface **28** may be the ground, one or more stands/frames that raise the nacelle assembly **24** off the ground, or other support surface. The handling location may be at the manufacturing facility, a loading facility, unloading facility, or any other type of intermediate or end-point facility in the chain from the manufacturing facility to the installation site where the handling of the nacelle **14** may be necessary.

(19) At the various handling locations, it may be necessary to move the nacelle **14** from one place to another. To this end, handling locations typically include one or more lifting apparatus **30** for hoisting the nacelle **14** off the support surface **28** and moving the nacelle **14** to another location. FIG. 3 illustrates an exemplary embodiment where the lifting apparatus **30** is configured as a dual-crane arrangement **32**. The dual-crane arrangement **32** includes a first crane **34** having a lifting cable **36** and a cable endpiece **38** configured to be attached to a front end of the nacelle **14**, and a second crane **40** having a lifting cable **36** and a cable endpiece **38** configured to be attached to a rear end of the nacelle **14**. In one aspect of the invention, and as will be discussed in more detail below, the endpieces **38** of the lifting cables **36** of the cranes **34**, **40** are configured to be attached to the transport frames **26** at the front and rear ends of the nacelle **14**. Connecting the lifting cables **36** to the transport frames **26** as opposed to the nacelle **14** itself, minimizes the risk of damaging the nacelle **14** during the movement of the nacelle **14** using the dual-crane arrangement **32**.

(20) In any event, the two cranes **34**, **40** may then be moved in a coordinated manner to move the nacelle assembly **24** from a first position or location to a second position or location at the handling location as known to those of ordinary skill in the art. While the lifting apparatus **30** illustrated in the figures includes a pair of cranes **34**, **40**, it should be understood that the lifting apparatus **30** may take a wide range of forms, including various mobile or stationary cranes, various permanent or temporary cranes, as well as various arrangements of winches, drums, lifting cables, booms,

counterweights, etc. that typically accompany lifting apparatus. Accordingly, the lifting apparatus should not be limited to any particular type or arrangement so long as the lifting apparatus has a lifting cable, an endpiece on the lifting cable, and the capability to hoist the load attached to the endpiece and move the load to another location.

(21) A transport frame **26** in accordance with an embodiment of the invention is shown in FIGS. **4** and **5** and is configured to connect to the front and rear ends of the nacelle **14**. The front and rear transport frames **26** are nearly identical to each other. Accordingly, only one transport frame **26** will be described in detail. It should be understood, however, that the other transport frame **26** has a similar construction and description. While some details of the transport frames **26** are provided below, additional details of the transport frames may be found in Applicant's application PCT/DK2020/050388. As illustrated in FIGS. **4** and **5**, the transport frame **26** has a multi-sectional design including a bottom frame section **44**, an intermediate frame section **46**, and a top frame section **48** coupled to each other to form the transport frame **26**.

(22) The bottom frame section **44** includes a base support **50**, which in an exemplary embodiment may take the form of a generally rectangular base plate, configured to support the transport frame **26** on the support surface **28**. Two generally vertical and spaced apart main beams **52**, **54** extend from an upper surface of the base support **50** with each terminating at an upper connecting end **56**. The base support **50** further includes a lower cross beam **58** that extends between and beyond the main beams **52**, **54** and which is generally parallel to the base support **50**. In one embodiment, the lower cross beam **58** may be a monolithic beam section that extends through openings formed in the main beams **52**, **54** and is welded or otherwise secured to the main beams **52**, **54**. The lower cross beam **58** includes an inboard cross beam section **60** between the two main beams **52**, **54** and a pair of outboard cross beam sections **62** on the outside of the main beams **52**, **54**. The outboard cross beam sections **62** of the cross beam **58** include lower nacelle mounts **64** configured to connect to lower nacelle tangs on the nacelle **14**.

(23) The intermediate frame section **46** includes two generally vertical and spaced apart main beams **66**, **68** with each terminating at a lower connecting end **70** and an upper connecting end **72**. The lower connecting ends **70** of the main beams **66**, **68** are configured to be coupled to the upper connecting ends **56** of the main beams **52**, **54** of the bottom frame section **44**. The main beams **66**, **68** of the intermediate frame section **46** may be coupled to the main beams **52**, **54** of the bottom frame section **44** by a hinged connection **74** that allows relative rotational movement between the intermediate frame section **46** and the bottom frame section **44**. The hinged connection **74** may further include one or more selectively engageable locks **76**.

(24) The top frame section **48** includes two generally vertical and spaced apart main beams **78**, **80** with each terminating at a lower connecting end **82**. The lower connecting ends **82** of the main beams **78**, **80** are configured to be coupled to the upper connecting ends **72** of the main beams **66**, **68** of the intermediate frame section **46**. The main beams **78**, **80** of the top frame section **48** may be coupled to the main beams **66**, **68** of the intermediate frame section **46** by a hinged connection **84** that allows relative rotational movement between the top frame section **48** and the intermediate frame section **46**. The hinged connection **84** may further include one or more selectively engageable locks **86**.

(25) The top frame section **48** further includes an upper cross beam **88** that extends between and beyond the main beams **78**, **80** and which is generally parallel to the base support **50** and lower cross beam **58**. In one embodiment, the upper cross beam **88** may be a monolithic beam section that extends through openings formed in the main beams **78**, **80** and are welded or otherwise secured to the main beams **78**, **80**. The upper cross beam **88** includes an inboard cross beam section **90** between the two main beams **78**, **80** and a pair of outboard cross beam sections **92** on the outside of the main beams **78**, **80**. The outboard cross beam sections **92** of the upper cross beam **88** includes upper nacelle mounts **94** configured to connect to upper nacelle tangs on the nacelle **14**.

(26) As illustrated in FIG. **3**, the endpieces **38** of the lifting cables **36** on the cranes **34**, **40** are

configured to engage with transport frames **26** at the front and rear ends of the nacelle **14**. In accordance with one aspect of the invention, the endpieces **38** of the cranes **34**, **40** are configured to interface with multiple wind turbine components being hoisted by the lifting apparatus **30**. As mentioned above, to hoist tower sections using the lifting apparatus **30**, the endpieces **38** may be configured as J-hooks. As previously noted, the challenge is to then use the J-hooks to also interface with the transport frames attached to the front and rear ends of the nacelle in a manner that avoids the balance and centre of gravity concerns that have limited the use of J-hooks for nacelle handling in the past. As discussed in more detail below, the inventor has discovered a solution for overcoming the balance and centre of gravity issues associated with using J-hooks to hoist nacelles. More particularly, the inventor has developed an adaptor that may be disposed between the J-hook and a transport frame that provides lateral adjustability in order to accommodate displacements of the centre of gravity of the nacelle assembly away from the geometric central plane between the two J-hooks coupled to the nacelle assembly. By providing this lateral adjustability, the nacelle assembly may be hoisted by the lifting apparatus in a balanced manner (i.e., without the nacelle tilting to one side or the other when lifted off the support surface).

(27) FIG. **6** illustrates an exemplary J-hook **100** that may be used according to aspects of the present invention. The J-hook **100** includes a plate-like body **102** defining opposed first and second surfaces or faces **104**, **106** and a sidewall **108** extending between the two faces **104**, **106**. The body **102** is generally J-shaped with a first leg **110** having upper end **112** and a second leg **114** extending from a lower end of the first leg **110**, such as at about 90 degrees, and terminating at an outer end **116**. The upper end **112** of the J-hook **100** includes a cable eyelet **118** for attaching to the lifting cable **36** of the lifting apparatus **30**. The J-hook **100** includes a mouth **120** for receiving a portion of the wind turbine component being hoisted by the lifting apparatus **30**. The mouth **120** may be defined at least in part by an L-shaped interface having a first interface portion or ridge and a second interface portion or ridge. In an exemplary embodiment, the first ridge may be formed at an inner edge of the first leg **110** and the second ridge may be formed at an upper edge of the second leg **114**. The first ridge may further include a first engagement plate **122** attached to, for example, the inner edge of the first leg **110**. The second ridge may similarly include a second engagement plate **124** attached to, for example, the upper edge of the second leg **114**. In an alternative embodiment, the second engagement plate **124** may be omitted.

(28) To secure the J-hook **100** to the wind turbine component being hoisted by the lifting apparatus **30**, such as a tower section or the nacelle assembly **24**, the J-hook includes a first lock jaw **128** configured to be releasably engaged with the first engagement plate **122** of the J-hook **100** and a second lock jaw **130** configured to be releasably engaged with the second engagement plate **124**. In an exemplary embodiment, the first engagement plate **122** may include a plurality of bores **132** arranged in two columns, for example, in the first engagement plate **122** and configured to receive fasteners **134** for fixedly attaching the first lock jaw **128** at a selected position along the first engagement plate **122**. The fasteners **134** may be threaded fasteners for being received in threaded bores **132**, or lock pins received in the bores **132**. The position of the first lock jaw **128** along the first engagement plate **122** will depend on how the wind turbine component is received within the mouth **120** of the J-hook **100**. In any event, in one embodiment, the first lock jaw **128** includes a U-shaped bracket body **136** configured to be slidably engaged with the first engagement plate **122** (such as over the upper end of the plate **122**) and a bearing surface **138** configured to engage with the wind turbine component being hoisted by the lifting apparatus **30**.

(29) In an exemplary embodiment, the second lock jaw **130** includes an L-shaped jaw body formed by a pair of spaced apart L-shaped plates **140** connected by a bearing pad **142** that defines a bearing surface **144**. The second lock jaw **130** is configured to be slidably attached to the J-hook **100**. In this regard, the second lock jaw **130** fits down over the upper edge of the second leg **114** such that the plates **140** of the second lock jaw **130** are disposed on the outside of the first and second faces **104**, **106** of the hook body **102** and a bottom edge of the bearing pad **142** engages the second

engagement plate **124**. To slidably couple the second lock jaw **130** to the hook body **102**, the second leg **114** includes an elongate slot **146** and the plates **140** of the second lock jaw **130** include one or more aligned apertures (two shown). One or more slide pins **148** are inserted into the one or more aligned apertures so as to extend through the elongate slot **146**. This arrangement secures the second lock jaw **130** to the hook body **102** but allows the lock jaw **130** to slide along the second engagement plate **124**. In this way, the second lock jaw **130** may be adjusted so as to engage with the wind turbine component being hoisted by the lifting apparatus **30**, such as at bearing surface **144**. To maintain the second lock jaw **130** in engagement with the wind turbine component, the J-hook **100** may include a threaded set screw **150** configured to be received within a threaded bore **152** in a lip **154** at the outer end **116** of the second leg **114** of the hook body **102**. The set screw **150** has an abutment end (not shown) configured to engage the rear surface of the bearing pad **142** to effectively clamp the wind turbine component to the J-hook **100**. The set screw **150** further includes a head **156** for turning the set screw **150**, thereby adjusting the clamping pressure on the wind turbine component.

(30) FIGS. 7A and 7B schematically illustrate the hoisting of a nacelle assembly **24** using current J-hooks, such as J-hooks **100** described above. As noted above, in many cases the centre of gravity **158** of the nacelle assembly **24** does not lie within the geometric central plane **160**, where, for example, the J-hooks **100** typically attach to the transport frames **26**, and along which the lifting cables **36** extend (see FIG. 7A). Instead, the centre of gravity **158** may be laterally displaced by an offset **162** from the geometric central plane **160** to one side thereof. Accordingly, when the nacelle assembly **24** is hoisted by the lifting apparatus **30**, e.g., the two cranes **34**, **40** of the dual crane arrangement **32**, the nacelle assembly **24** tilts under the influence of gravity until the centre of gravity **158** lies vertically below the J-hooks **100** and aligns with the lifting cables **36**, as schematically illustrated in FIG. 7B. Manufacturers and other technicians that handle wind turbine components want to avoid such an imbalance in the load hoisted by the lifting apparatus **30**. Accordingly and as discussed above, the use of J-hooks **100** as the endpiece **38** of the lifting cables **36** is typically avoided when handling nacelles **16** or nacelle assemblies **24**, and a change-out procedure is necessary to provide a more suitable endpiece **38** to the lifting apparatus **30** for handling nacelles **14**.

(31) In accordance with an aspect of the invention, to address this deficiency, an adaptor **166** may be disposed between the J-hook **100** of the lifting apparatus **30** and the transport frame **26** of the nacelle assembly **24** that allows the centre of gravity **158** of the nacelle assembly **24** to be shifted laterally so that the centre of gravity **158** is vertically aligned with the eyelet **118** of the J-hook **100** (and more particularly the lifting cable **36** attached to the eyelet **118**). By providing this ability to adjust the lateral position of the nacelle assembly **24** relative to the J-hook **100**, when the nacelle assembly **24** is lifted by the lifting apparatus **30**, tilting of the nacelle assembly **24** is avoided and a balanced movement of the nacelle assembly **24** from one position to another using the lifting apparatus **30** may be achieved. Accordingly, the J-hooks **100**, which are traditionally used to move tower sections, may also be used to move nacelle assemblies **24** at a handling location with the addition of adaptors **166** to the J-hooks **100**. As explained in more detail below, the addition of the adaptors **166** to the J-hooks **100** is a relatively easy and quick step. Accordingly, the change-out processes currently used for handling multiple wind turbine components is avoided and moving wind turbine components at a handling location becomes more manageable and time efficient.

(32) In one embodiment, and as described below, the adaptor **166** may be configured to be coupled to the J-hook **100**, thereby forming a J-hook assembly that is then subsequently coupled to the transport frames **26** of the nacelle assembly **24**. In other words, the adaptor **166** essentially becomes a part of the J-hook **100** for use with handling the nacelle assembly **24**. However, in an alternative embodiment, the adaptor **166** may be configured to be coupled to the transport frames **26**, thereby forming transport frame assemblies that are subsequently coupled to traditional types of J-hooks **100**, such as that shown in FIG. 6. In either embodiment, the adaptor **166** is disposed between the

J-hook **100** and the transport frames **26** to provide lateral adjustments of the nacelle assembly **24** to vertically align the offset centre of gravity **158** of the nacelle assembly **24** with the eyelet **118** of the J-hook **100** and the lifting cable **36** of the lifting apparatus **30**.

(33) FIGS. **8-8D** illustrate an adaptor **166** in accordance with an exemplary embodiment of the invention. The adaptor **166** includes an adaptor body **168** including a bracket portion **170** for attaching to the J-hook **100** and a projection portion **172** for attaching to the transport frame **26**. The adaptor body **168** may be formed from metal, such as steel, and be formed as an integral body or as separate elements coupled together, such as by welding, to form the adaptor body **168**. Other materials may also be possible. The adaptor body **168** includes an upper base plate **174** and two side legs **176**, **178** extending from the base plate **174**. In one embodiment, the side legs **176**, **178** extend from the base plate **174** at about 90 degrees (i.e., perpendicularly). However, the side legs **176**, **178** may extend from the base plate **174** at other angles in alternative embodiments. In an exemplary embodiment, the base plate **174** and side legs **176**, **178** may have a generally rectangular cross-sectional profile, but may have other cross-sectional profiles in alternative embodiments, such as circular, oval, square, triangular, or other polygonal profiles. In one embodiment, the base plate **174** may include an outer surface **180**, an opposed inner surface **182**, a front surface **184**, an opposed rear surface **186**, and opposed side surfaces **188**, **190**. The side legs **176**, **178** extend from the inner surface **182** of the base plate **174** and include an outer surface **192**, an opposed inner surface **194**, a front surface **196**, an opposed rear surface **198**, and an end surface **200** opposed to where the legs **176**, **178** attach to the base plate **174**. The combination of the base plate **174** and side legs **176**, **178** forms a generally inverted U-shaped recess **202**.

(34) In one embodiment (not shown), the side legs **176**, **178**, may be coextensive with the base plate **174** in length from the front surface **184** to the rear surface **186**. In an alternative embodiment, however, and as illustrated in FIGS. **8-8D**, the side legs **176**, **178** may have a length less than the length of the base plate **174**. Moreover, in an exemplary embodiment the side legs **176**, **178** may extend from base plate **174** adjacent the rear surface **186** of the base plate **174**. In alternative embodiments, however, the side legs **176**, **178** may extend from the base plate **174** at other locations between the front and rear surfaces **184**, **186**. In an exemplary embodiment, the side legs **176**, **178** are generally aligned with each other across the base plate **174**, as illustrated in FIGS. **8-8D**. Aspects of the adaptor **166**, however, are not so limited and the side legs **176**, **178** may be spaced apart or offset from each other in the length direction of the base plate **174**.

(35) As illustrated in the figures, the outer surface **180** of the base plate **174** includes one or more locating projections **204** (two shown) extending away from the surface **180**. As will be explained in more detail below, the projections **204** facilitate a connection between the adaptor **166** and the transport frame **26** at a predetermined location of the transport frame **26**. The projections **204** may be configured as pegs, posts, fingers, tabs, etc. extending from the outer surface **180** of the base plate **174** of the adaptor **166**. The projections **204** may have a wide range of lengths and cross-sectional profiles. For example, each of the one or more projections **204** may include a generally cylindrical post having a circular cross-sectional profile. In an exemplary embodiment, a first, lower portion of the projections **204** adjacent the outer surface **180** may include a generally constant diameter section and a second, outer portion of the projections **204** may include a generally conical section that terminates at a tip **206**. The tapered configuration of the projections **204** guides the projections **204** into complementary recesses or apertures in the transport frames **26**, as will be discussed in more detail below. The projections **204**, however, are not limited to this configuration and may have other configurations for engaging with complementary receivers in the transport frames **26**.

(36) As further shown in FIGS. **8-8D**, to provide lateral adjustability, the adaptor **166** further includes a pair of set screws **208**, **210** associated with respective side legs **176**, **178**. More particularly, each of the side legs **176**, **178** includes a threaded bore **212** extending between the outer surface **192** and the inner surface **194** and configured to receive respective set screws **208**,

210 therein. In an exemplary embodiment, the threaded bores **212** may be generally parallel to the base plate **174** and be positioned along the side legs **176, 178** adjacent the base plate **174**. By way of example, the threaded bores **212** may be immediately adjacent the inner surface **182** of the base plate **174**. However, positions slightly spaced from the inner surface **182** may also be possible. The set screws **208, 210** include a threaded shaft **214**, an enlarged head **216**, and an abutment end **218** configured to engage with the J-hook **100** in a manner described below. The enlarged heads **216** are outboard of the outer surfaces **192** of the side legs **176, 178** and the abutments ends **218** are positioned in the U-shaped recess **202** between the side legs **176, 178**. As is readily understood, rotation of the set screws **208, 210** moves the abutment ends **218** toward and away from each other. (37) FIG. **9** illustrates the adaptor **166** coupled to the J-hook **100** to form a J-hook assembly **222** for mating to the transport frames **26** of the nacelle assembly **24**. In this regard, the adaptor **166** is placed over the second engagement plate **124** and disposed between the first engagement plate **122** and the second lock jaw **130** such that the inner surface **182** of the base plate **174** engages with the second engagement plate **124** of the J-hook **100**. In this position, the side legs **176, 178** project downwardly from the base plate **174** such that the inner surfaces **194** of the side legs **176, 178** confront the first and second faces **104, 106** of the second leg **114** of the J-hook **100**. The distance between the inner surfaces **194** of the side legs **176, 178** (i.e., the width of recess **202**) is greater than the width of the second leg **114** of the J-hook **100** (i.e., the distance between the first and second faces **104, 106** of the J-hook **100** along the second leg **114**). In this way, the adaptor **166** is movable laterally relative to the J-hook **100**. This lateral movement is bounded on one side by the engagement of the first side leg **176** with the first face **104** of the J-hook **100** and bounded on the other side by the engagement of the second side leg **178** with the second face **106** of the J-hook **100**. If the centre of gravity **158** of the nacelle assembly **24** is within this lateral range of the adaptor **166** on the J-hook **100**, then the lifting apparatus **30** will be able to hoist the nacelle assembly **24** in a balanced configuration.

(38) To this end, the goal is to align the eyelet **118** of the J-hook **100**, and more particularly the lifting cable **36** of the lifting apparatus **30**, with the laterally offset centre of gravity **158** of the nacelle assembly **24**. If the lateral offset **62** of the centre of gravity **158** of the nacelle assembly **24** is known, then the adaptor **166** may be secured to the J-hook **100** in an asymmetric manner (that is with the adaptor **166** shifted laterally to one side or the other of the second leg **114** of the J-hook **100**) such that the centre of gravity **158** of the nacelle assembly **24** is vertically aligned with the eyelet **118** of the J-hook **100**. In this regard, once the adaptor **166** is offset by the appropriate amount, the first and second set screws **208, 210** may be tightened until the abutment ends **218** engage the respective faces **104, 106** of the J-hook **100**. The second lock jaw **130** of the J-hook **100** may then be moved into position such that the bearing surface **144** of the bearing pad **142** engages the front surface **184** of the base plate **174** of the adaptor **166** to clamp the adaptor **166** against the first engagement plate **122** at the rear surface **186** of the base plate **174**. However, the clamping of the adaptor **166** with the second lock jaw **130** may be optional. In any event, between the set screws **208, 210** and possibly the second lock jaw **130**, the adaptor **166** may be securely attached to the J-hook **100** in the desired offset configuration.

(39) FIGS. **10A** and **10B** illustrate the use of the J-hook assembly **222** in handling a nacelle assembly **24**. To this end and as illustrated in these figures, the transport frames **26** include a connection interface **224** (FIG. **4**) for mating with the adaptor **166** on the J-hook assembly **222**. In an exemplary embodiment, the connection interface **224** includes one or more face plates **226** attached to the underside **228** of the upper cross beam **88** of the transport frames **26**. The face plates **226** may be made of metal, such as steel, and either welded or bolted to the upper cross beam **88**. The face plates **226**, and possibly the upper cross beam **88**, includes one or more apertures **230** configured to receive the one or more projections **204** on the J-hook assembly **222**. To this end, the lifting apparatus **30**, such as one of the cranes **34, 40** may maneuver the J-hook assembly **222** outside of and adjacent to the transport frame **26** such that the projections **204** of the adaptor **166**

are just below the upper cross beam **88** of the transport frame **26**. This arrangement is illustrated in FIG. **10A**. Service personnel on the ground may then position the J-hook assembly **222** such that the projections **204** engage with the apertures **230** in the connection interface **224** as the J-hook assembly **222** is slightly raised by the lifting apparatus **30**, thereby attaching the J-hook assembly **222** to the transport frame **26**. The same process may be used to attach the other J-hook assembly **222** to the other transport frame **26** on the other side of the nacelle assembly **24**. This arrangement is illustrated in FIG. **10B**.

(40) It is noted that the connection between the J-hook assembly **222** and the transport frame **26** is tool-less, which provides certain benefits compared to current methods. In any event, after connecting the J-hook assemblies **222** to the transport frames **26**, the nacelle assembly **24** may then be moved from one location to another location using the lifting apparatus **30**. At the new location, the J-hook assemblies **222** may be disengaged from the transport frames **26** by lowering the J-hook assemblies **222** so that the projections **204** disengage the apertures **230**.

(41) FIG. **11** illustrates the adaptor **166** in a laterally offset configuration relative to the J-hook **100** such that the centre of gravity **158** of the nacelle assembly **24** is vertically aligned with the eyelet **118** of the J-hook **100** and the lifting cable **36**. Accordingly, when the nacelle assembly **24** is lifted off the support surface **28**, such as the ground or stand, the nacelle assembly **24** does not tilt or otherwise rotate. The ability to adjust the lateral position of the adaptor **166** on the J-hook **100** is what allows the nacelle assembly **24** to be hoisted by the lifting apparatus **30** in a balanced configuration that avoids the tilting of the nacelle assembly **24** during the lift. As noted above, if the lateral offset of the centre of gravity **158** of the nacelle assembly **24** is known, then the adaptors **166** may be arranged on the J-hooks **100** to correspond to the lateral offset and achieve a balanced lift.

(42) If, on the other hand, the centre of gravity **158** of the nacelle assembly **24** is not precisely known, an initial offset position of the adaptor **166** on the J-hooks **100** may be made. The lifting apparatus **30** may then lift the nacelle assembly **24** to determine whether the assembly **24** starts to tilt upon being lifted off the support surface **28**. FIG. **12**, for example, illustrates a scenario where the initial offset for the adaptor **166** relative to the J-hook **100** is selected to be zero (i.e., the adaptor **166** is centrally positioned on the second engagement plate **124** of the J-hook **100**) and, as a consequence, the nacelle assembly **24** tilts as the assembly **24** is being lifted by the lifting apparatus **30**. In this case, the nacelle assembly **24** may be lowered back to the support surface **28**, the adaptors **166** adjusted to change the initial lateral offset value relative to the J-hooks **100** (i.e., moving the adaptors **166** laterally in one direction or the other depending on which way the nacelle assembly **24** tilts), and the nacelle assembly **24** raised again with the lifting apparatus **30**. This iterative or trial-and-error process may be repeated until the appropriate lateral offset is found that vertically aligns the centre of gravity **158** of the nacelle assembly **24** and the eyelet **188** of the J-hooks **100** and/or the lifting cables **36** of the lifting apparatus **30** and tilting of the nacelle assembly **24** is eliminated upon being hoisted by the lifting apparatus **30**.

(43) While the present invention has been illustrated by a description of various preferred embodiments and while these embodiments have been described in some detail, it is not the intention of the Applicant to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. Thus, the various features of the invention may be used alone or in any combination depending on the needs and preferences of the user.

Claims

1. An adaptor for interfacing between an endpiece of a lifting apparatus and a nacelle of a wind turbine, comprising: a base plate having an outer surface and an inner surface, the outer surface having one or more projections configured to engage the adaptor to the nacelle; a first side leg

extending away from the inner surface of the base plate, wherein the first side leg includes an outer surface, an inner surface, and a first bore extending between the inner and outer surfaces of the first side leg; a second side leg extending away from the inner surface of the base plate and spaced from the first side leg, wherein the second side leg includes an outer surface, an inner surface, and a second bore extending between the inner and outer surfaces of the second side leg, and wherein the base plate, first side leg, and second side leg collectively define a recess configured to engage the adaptor with the endpiece of the lifting apparatus; a first set screw positionable in the first bore such that an abutment end of the first set screw is disposed within the recess; and a second set screw positionable in the second bore such that an abutment end of the second set screw is disposed within the recess, wherein the first and second set screws are configured to be movable, relative to the first and second bores, respectively, toward and away from each other to adjustably position the adaptor relative to the endpiece of the lifting apparatus.

2. The adaptor of claim 1, wherein the base plate is generally rectangular and further comprises a front surface, a rear surface, and opposed side surfaces, and wherein the first and second side legs extend from the inner surface adjacent respective side surfaces of the base plate.

3. The adaptor of claim 1, wherein the first and second side legs extend from the inner surface adjacent to the rear surface of the base plate.

4. The adaptor of claim 1, wherein the first and second side legs have a length that is less than a length of the base plate.

5. The adaptor of claim 1, wherein the first and second side legs are aligned across a width of the base plate.

6. The adaptor of claim 1, wherein the first and second side legs extend from the inner surface of the base plate at a substantially right angle.

7. The adaptor of claim 1, wherein the first and second set screws and the first and second bores are threaded such that rotation moves the first and second set screws toward and away from each other.

8. The adaptor of claim 1, wherein the first bore and the second bore extend through their respective side legs in a direction parallel to the base plate.

9. The adaptor of claim 1, wherein the first bore and the second bore extend through their respective side legs adjacent the inner surface of the base plate.

10. The adaptor of claim 1, wherein the first and second set screws are movable toward and away from each other along an axis, and wherein the adaptor is configured to be adjustable relative to the endpiece of the lifting apparatus in a direction parallel to the axis.

11. The adaptor of claim 1, wherein the one or more projections further comprises a plurality of posts spaced apart on the upper surface of the base plate, and wherein each of the plurality of posts has a tapered end.

12. The adaptor of claim 1, wherein the adaptor is configured to mount to the nacelle in a tool-less manner.

13. An endpiece assembly for a lifting apparatus having a lifting cable, comprising: an endpiece configured to be coupled to the lifting cable of the lifting apparatus, wherein the endpiece defines a lateral direction; and the adaptor of claim 1 mounted to the endpiece, wherein the position of the adaptor relative to the endpiece is adjustable in the lateral direction, and wherein the adaptor is coupled to the endpiece at least by the first and second set screws.

14. The endpiece assembly of claim 13, wherein the endpiece includes a J-hook.

15. The endpiece assembly of claim 14, wherein the J-hook further comprises: a first leg having an eyelet for attaching to the lifting cable and a first edge; a second leg extending from the first leg and having a second edge, wherein the adaptor is mounted on the second edge such that the inner surface of the base plate engages the second edge and the second leg of the J-hook is disposed in the recess between first and second side legs of the adaptor.

16. The endpiece assembly of claim 15, wherein the distance between the inner surfaces of the first and second side legs of the adaptor is greater than a thickness of the second leg of the J-hook.

17. The endpiece assembly of claim 15, wherein the J-hook includes a lock jaw for clamping the adaptor to the J-hook.

18. A method of handling wind turbine components at a facility, comprising: providing a lifting apparatus having a lifting cable and the endpiece assembly of claim 13 on the lifting cable for interfacing with a load for lifting with the lifting apparatus; providing a first wind turbine component for moving with the lifting apparatus, the first wind turbine component having a centre of gravity laterally offset from a central plane of the first wind turbine component; selectively disposing the adaptor between the endpiece of the lifting apparatus and the first wind turbine component, the adaptor being adjustable relative to the endpiece of the lifting apparatus in the lateral direction; positioning the adaptor relative to the endpiece of the lifting apparatus so that the endpiece is vertically aligned with the offset centre of gravity of the first wind turbine component; and lifting the first wind turbine component with the lifting apparatus.

19. The method of claim 18, wherein positioning the adaptor relative to the endpiece of the lifting apparatus further comprises: predetermining the lateral offset of the centre of gravity of the first wind turbine component; and prior to lifting the first wind turbine component, adjusting the position of the adaptor relative to the endpiece of the lifting apparatus to correspond to the predetermined lateral offset.

20. The method of claim 18, wherein positioning the adaptor relative to the endpiece of the lifting apparatus further comprises iteratively adjusting the position of the adaptor relative to the endpiece of the lifting apparatus and lifting the first wind turbine component with the lifting apparatus until the first wind turbine component fails to tilt when lifted by the lifting apparatus.

21. The method of claim 18, further comprising using the endpiece of the lifting apparatus without the adaptor to lift a second wind turbine component at the facility.

22. The method of claim 21, wherein the second wind turbine component is a tower section.

23. The method of claim 18, wherein the first wind turbine component includes a nacelle assembly, comprising: a nacelle having a front end and a rear end; a first transport frame attached to the front end of the nacelle; and a second transport frame attached to the rear end of the nacelle.

24. The method of claim 23, wherein the lifting apparatus comprises a dual-crane system, comprising: a first crane having a first lifting cable and a first endpiece; and a second crane having a second lifting cable and a second endpiece, wherein the method further comprises: selectively disposing a first adaptor between the first endpiece and the first transport frame of the nacelle assembly; positioning the first adaptor relative to the first endpiece of the lifting apparatus so that the first endpiece is vertically aligned with the offset centre of gravity of the nacelle assembly; selectively disposing a second adaptor between the second endpiece and the second transport frame of the nacelle assembly; positioning the second adaptor relative to the second endpiece of the lifting apparatus so that the second endpiece is vertically aligned with the offset centre of gravity of the nacelle assembly; and lifting the nacelle assembly with the dual crane system.

25. The method of claim 18, wherein the endpiece includes a J-hook.
